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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2025

AD1101-1 – COMPUTER ORGANIZATION
 (For AID)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which of the following is a functional unit of a computer system?
 A) Arithmetic and Logic Unit (ALU) B) Operating System
 C) Keyboard D) Monitor
2. What is the sequence of steps followed by a CPU in executing instructions known as?
 A) Instruction Cycle B) Machine Cycle
 C) Execution Cycle D) Fetch-Decode-Execute Cycle
3. What is computer organization?
 A) structure and behaviour of a computer system as observed by the user B) structure of a computer system as observed by the developer
 C) structure and behaviour of a computer system as observed by the developer D) structure of a computer system
4. What is the main function of the Control Unit in a processor?
 A) Control Unit B) ALU
 C) Memory D) Cache
5. Which equation represents the basic performance equation?
 A) Performance = (Clock Rate) / (CPI × Instruction Count) B) Performance = $\frac{\text{Clock Rate}}{\text{CPU Time}}$
 C) Performance = (Instruction Count × CPI) / Clock Rate D) Performance = Clock Rate × Instruction Count × CPI
6. Which of the following directly affects a processor's clock speed?
 A) Instruction Set Architecture B) Number of transistors
 C) Length of the clock cycle D) Cache size
7. What is meant by the term "throughput" in performance measurement?
 A) The number of instructions executed per second B) The number of processes handled by the CPU
 C) The speed at which the CPU fetches instructions D) The number of instructions processed in parallel

8. How are multiple I/O devices prioritized in an interrupt-driven system?
- A) Through the CPU clock
B) Using an interrupt priority scheme
C) Polling each device sequentially
D) Using DMA
9. Which of the following is an exception?
- A) A type of interrupt caused by software error or specific event
B) A signal from an I/O device requesting CPU attention
C) A hardware fault occurring in the CPU
D) An increase in the CPU clock speed
10. What is Direct Memory Access (DMA) primarily used for?
- A) Handling arithmetic operations
B) Generating interrupts for the CPU
C) Handling data transfers between I/O devices and memory without CPU intervention
D) Managing multiple processors
11. In a bus system, what is the main function of the bus?
- A) To transfer data between different components of a computer system
B) To decrease system speed.
C) To increase CPU speed
D) To manage interrupts
12. Which type of bus is used to connect high-speed peripherals to the CPU?
- A) Data bus
B) Address bus
C) PCI bus
D) Control bus
13. The PCI bus operates primarily in:
- A) Parallel
B) Serial
C) Combination of parallel and serial
D) Only synchronous mode
14. Which of the following is a key feature of USB?
- A) Serial data transfer
B) Parallel data transfer
C) Direct memory access
D) Handles interrupts directly
15. Which of the following types of memory is volatile?
- A) ROM
B) RAM
C) Flash memory
D) PROM
16. What does DRAM stand for?
- A) Direct Random Access Memory
B) Dynamic Random Access Memory
C) Delayed Read Access Memory
D) Discrete Random Access Memory
17. SRAM is typically used for:
- A) Main memory
B) External storage
C) Permanent storage
D) Cache memory
18. Which type of ROM can be erased and rewritten multiple times?
- A) PROM
B) EPROM
C) EEPROM
D) Masked ROM
19. Which type of memory has the highest cost per bit?
- A) DRAM
B) SRAM
C) ROM
D) Magnetic disk
20. What is the purpose of cache memory in a computer system?
- A) To store frequently accessed data for faster access
B) To provide permanent storage
C) To reduce the cost of memory
D) To store system configuration data

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I**

	Marks	BT*	CO*	PO*
1. a) Explain to measure the performance of a computer? Explain.	8	L*2	1	1
b) With a neat diagram, explain the basic operational concepts of a computer.	8	L2	1	1
2. a) Solve the following: i). $(110011)_2$ to Decimal ii). $(89DF)_{16}$ to Binary iii). $(1000)_{10}$ to Binary iv). $(101101)_2$ to Octal	8	L3	2	1
b) Explain restoring binary division algorithm, Perform 13/3 using restoring binary division algorithm.	8	L2	2	1
3. a) With examples, explain various addressing modes.	8	L3	2	1
b) Explain the concept of Subroutine? Differentiate between Queue and Stack.	8	L3	2	1

Unit – II

4. a) Explain Booth algorithm and apply booth algorithm to multiply signed numbers 7 and 6.	8	L3	3	1
b) Explain the algorithm of restoring for signed numbers and non-restoring division for unsigned numbers.	8	L2	3	1
5. a) Define Bus Arbitration. Explain in detail any one approach of bus arbitration.	8	L2	4	1
b) With suitable diagram, explain the following with respect to interrupts: i) Interrupt Nesting ii) Simultaneous Request	8	L2	4	1
6. a) Illustrate DMA controller with a neat diagram.	8	L2	3	1
b) Explain the concept of synchronous bus and PCI bus.	8	L2	4	1

Unit – III

7. a) Compare SRAM, DRAM, and Flash memory in terms of structure, speed, cost, and applications.	8	L2	5	1
b) Explain the concept of Cache memory.	8	L2	5	1
8. a) What is ROM memory? What are the different types of ROM. Explain?	8	L2	5	1
b) Explain the concept of virtual memory.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
